

# Comparing Unlike Fractions in Real Situations

## Answer Key

Grade 4–5 Mathematics

### Quick Answers

1. —

2.  $\frac{3}{4}$ 

3. —

4. —

5.  $\frac{7}{10}$ 

6. —

7.  $\frac{1}{2}, \frac{3}{5}, \frac{7}{10}$ 

8. —

9. —

10.  $\frac{3}{4}, \frac{2}{3}, \frac{5}{8}, \frac{7}{12}$ 

**1. Ana  $\frac{2}{5}$  of a bottle vs. Ben  $\frac{3}{4}$  of the same bottle.** Half of five fifths is  $\frac{2.5}{5}$ , so  $\frac{2}{5}$  is *less* than  $\frac{1}{2}$ . Half of four fourths is  $\frac{2}{4}$ , and  $\frac{3}{4}$  is more than that, so  $\frac{3}{4}$  is *greater* than  $\frac{1}{2}$ . One fraction sits below the benchmark and the other above it, so no common denominator is needed. Ben has more water.  $\frac{2}{5} < \frac{3}{4}$

**2.  $\frac{3}{4}$  of a tray written in twelfths.** The tray is cut into 12 equal pieces, so the new denominator is 12. Since  $4 \times 3 = 12$ , multiply the numerator by the same factor:  $\frac{3}{4} = \frac{3 \times 3}{4 \times 3}$ . Dev took 9 of the 12 pieces.  $\frac{9}{12}$

*Why multiplying top and bottom is legal:*  $\frac{3}{3} = 1$ , and multiplying by 1 cuts each piece into smaller pieces without changing how much cornbread is on the plate.

**3. Sam  $\frac{2}{3}$  vs. Kim  $\frac{5}{8}$ .** Both are just above  $\frac{1}{2}$ , so the benchmark cannot separate them — use the common denominator 24 ( $3 \times 8 = 24$ ).

$$\frac{2}{3} = \frac{2 \times 8}{3 \times 8} = \frac{16}{24},$$

$$\frac{5}{8} = \frac{5 \times 3}{8 \times 3} = \frac{15}{24}$$

With the same size pieces, 16 pieces beat 15 pieces. Sam walked farther.  $\frac{2}{3} > \frac{5}{8}$

**4. Morning crew  $\frac{4}{9}$  vs. afternoon crew  $\frac{5}{6}$ .** Half of nine ninths is  $\frac{4.5}{9}$ , so  $\frac{4}{9}$  falls just short of  $\frac{1}{2}$ . Half of six sixths is  $\frac{3}{6}$ , and  $\frac{5}{6}$  is well past it. Below the benchmark is always less than above it.  $\frac{4}{9} < \frac{5}{6}$

**5.  $\frac{7}{10}$  of a cup written in fortieths.** The target denominator is 40 and  $10 \times 4 = 40$ , so multiply the numerator by 4 as well:  $\frac{7}{10} = \frac{7 \times 4}{10 \times 4}$ .  $\frac{28}{40}$

**6. Mia  $\frac{5}{6}$  vs. Jonah  $\frac{7}{9}$ .** Both are above  $\frac{1}{2}$ , so build a common denominator. The least common multiple of 6 and 9 is 18.

$$\frac{5}{6} = \frac{5 \times 3}{6 \times 3} = \frac{15}{18}, \quad \frac{7}{9} = \frac{7 \times 2}{9 \times 2} = \frac{14}{18}$$

15 eighteenths is more than 14 eighteenths, so Mia has read more.  $\boxed{\frac{5}{6} > \frac{7}{9}}$

**7. Order  $\frac{3}{5}$ ,  $\frac{1}{2}$ ,  $\frac{7}{10}$  from least to greatest.** Every denominator divides 10, so tenths is the common unit:  $\frac{3}{5} = \frac{6}{10}$ ,  $\frac{1}{2} = \frac{5}{10}$ , and  $\frac{7}{10}$  already has it. Order the numerators  $5 < 6 < 7$ , then rewrite the fractions in their original form. The Baker family planted least, the Chen family most.

$\boxed{\frac{1}{2} < \frac{3}{5} < \frac{7}{10}}$

**8.  $\frac{7}{8}$  full vs.  $\frac{9}{10}$  full.** Both are above  $\frac{1}{2}$ , so measure the missing part instead: the first tank is  $\frac{1}{8}$  from full, the second is  $\frac{1}{10}$  from full. With the same numerator, the fraction with the *larger* denominator is the smaller amount, so  $\frac{1}{10} < \frac{1}{8}$ : the second tank is missing less.

Missing less means holding more.  $\boxed{\frac{7}{8} < \frac{9}{10}}$

*Check with a common denominator:*  $\frac{7}{8} = \frac{35}{40}$  and  $\frac{9}{10} = \frac{36}{40}$ , and  $35 < 36$ . Same conclusion.

**9. Recipe A  $\frac{11}{15}$  vs. Recipe B  $\frac{3}{4}$ .** Both sit between  $\frac{1}{2}$  and 1, so the benchmark is useless here. The least common multiple of 15 and 4 is 60.

$$\frac{11}{15} = \frac{11 \times 4}{15 \times 4} = \frac{44}{60}, \quad \frac{3}{4} = \frac{3 \times 15}{4 \times 15} = \frac{45}{60}$$

The recipes differ by only  $\frac{1}{60}$  of a cup, and Recipe B is the larger.  $\boxed{\frac{11}{15} < \frac{3}{4}}$

**10. Order  $\frac{7}{12}$ ,  $\frac{5}{8}$ ,  $\frac{2}{3}$ ,  $\frac{3}{4}$  from greatest to least.** All four denominators divide 24, so 24 is the common unit.

$$\frac{7}{12} = \frac{14}{24}, \quad \frac{5}{8} = \frac{15}{24}, \quad \frac{2}{3} = \frac{16}{24}, \quad \frac{3}{4} = \frac{18}{24}$$

Ordering the numerators from greatest to least gives  $18 > 16 > 15 > 14$ , so Nia walked the farthest and Rosa the least.  $\boxed{\frac{3}{4} > \frac{2}{3} > \frac{5}{8} > \frac{7}{12}}$

*Grading note:* accept any correct common denominator (a student who uses 48 or 72 gets the same order). The order is what is being assessed.